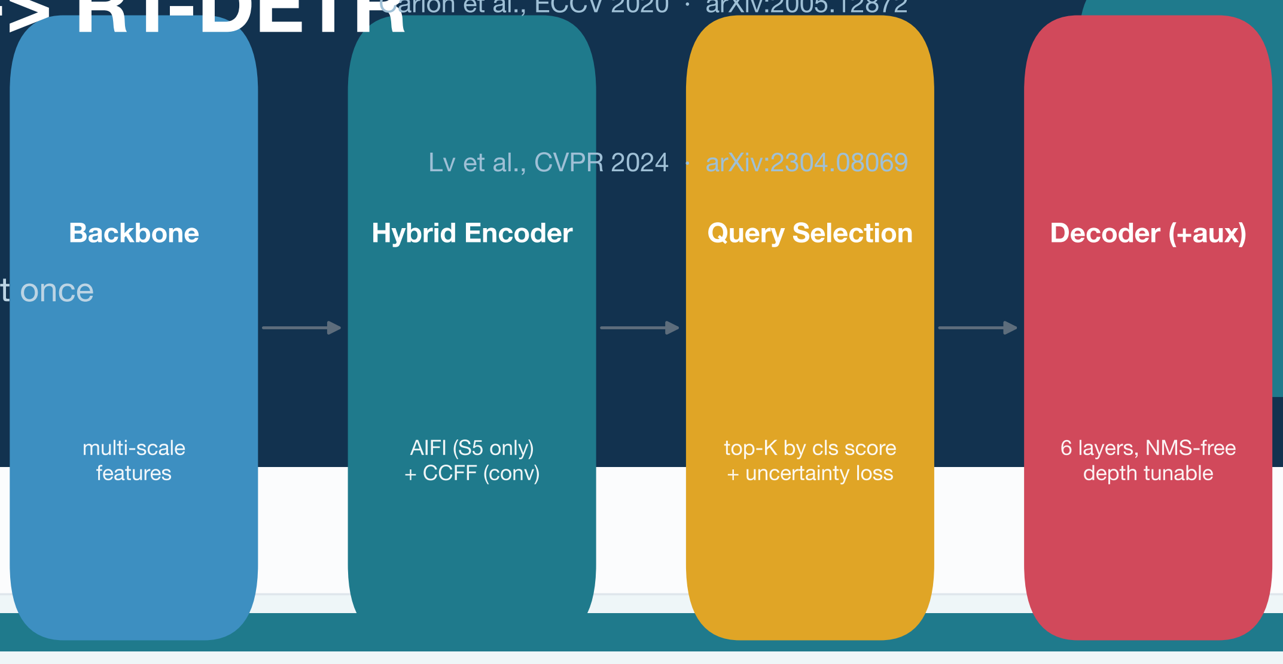
NMS-Free Real-Time Detection: DETR -> RT-DETR

arion et al., ECCV 2020 · arXiv:2005.12872

Lv et al., CVPR 2024 · arXiv:2304.08069

Why moving duplicate suppression from post-processing into the loss buys accuracy and speed at once



Backbone

multi-scale features

Hybrid Encoder

AIFI (S5 only) + CCFF (conv)

Query Selection

top-K by cls score + uncertainty loss

Decoder (+aux)

6 layers, NMS-free depth tunable

42.0

DETR AP (41M)

53.1

RT-DETR AP

108

FPS on T4 (FP16)

Model	AP	AP_S	AP_L	Params	FPS
DETR (R50)	42.0	20.5	61.1	41M	28
Faster R-CNN-FPN+	42.0	26.6	53.4	42M	26
RT-DETR (R50)	**53.1**	**34.8**	**70.0**	**42M**	**108**

DETR: same params as Faster R-CNN-FPN+, but AP_S -5.5 / AP_L +7.8 | RT-DETR: +2.2 AP and 21x faster than DINO-Deformable (5 -> 10

1 The NMS dilemma

- Two thresholds move accuracy and time in opposite directions
- conf 0.001 -> 0.05: AP 52.9 -> 51.2, NMS 2.36 -> 1.06 ms
- IoU 0.5 -> 0.8: AP 52.1 -> 52.8, NMS 2.24 -> 2.46 ms

Fig. 1 NMS cost model (our measurement, direction matches paper Table 1)

2 Three moves

One-to-one matching kills duplicates at training time -> no NMS at inference

8400 vs 400 tokens = 441x attention gap -> keep attention only on S5

- 1) Duplicates -> loss: optimal bipartite matching (one slot per GT)
- matching cost = probability, loss = log-likelihood
- box loss = scale-invariant GIoU + L1 (weights 2 / 5)
- 2) Compute -> design: encoder was 49% of GFLOPs, 11% of AP
- AIFI (S5 only, 400 tokens) + CCFF (conv fusion)

Accuracy and speed together

Conditional comparison: 640x640 / 72 ep (RT-DETR) vs 800x1333 / 500 ep (DETR)

Fig. 2 Accuracy vs speed (paper Table 2): RT-DETR is top-right

4 Our own experiments

- YOLO papers report model speed, excluding NMS time

L1 scales with size, GIoU does not (9.44x vs 1.000x)

Attention cost ~ M^2 (42.3x for 8x tokens)

Encoder variant latency order matches the paper

Uncertainty alone has a degenerate optimum

5 Limitations we kept honest

- Small objects still weak: AP_S 34.8 (-0.5 vs YOLOv8-L; R101 -0.9)
- 49% GFLOPs / 11% AP is a second-hand statistic, not our measurement
- 138% / 120% are relative counts, not percentage-point gains

6 Study trail and what is next

D1 MAE

D2 BEiT

D3 iBOT

D4-6 DeepSeek

D7 DETR

D8 RT-DETR

- 16 deep-dive docs, 4 runnable demos, 22 charts, 0 citation errors
- Next: GLIP + SAM (open-vocab detection -> prompt segmentation)
- Then: CLIP -> LLaVA (2 days) -> VL-JEPA | parallel 3D track

arXiv:2005.12872 · arXiv:2304.08069 · code: github.com/facebookresearch/detr, github.com/lyuwenyu/rt-detr · figures: our Day 07-08 measurements (synthetic, CPU)

Move duplicate removal into the loss, move compute into the design, move initialization into quality.

set prediction

hybrid encoder

uncertainty-minimal